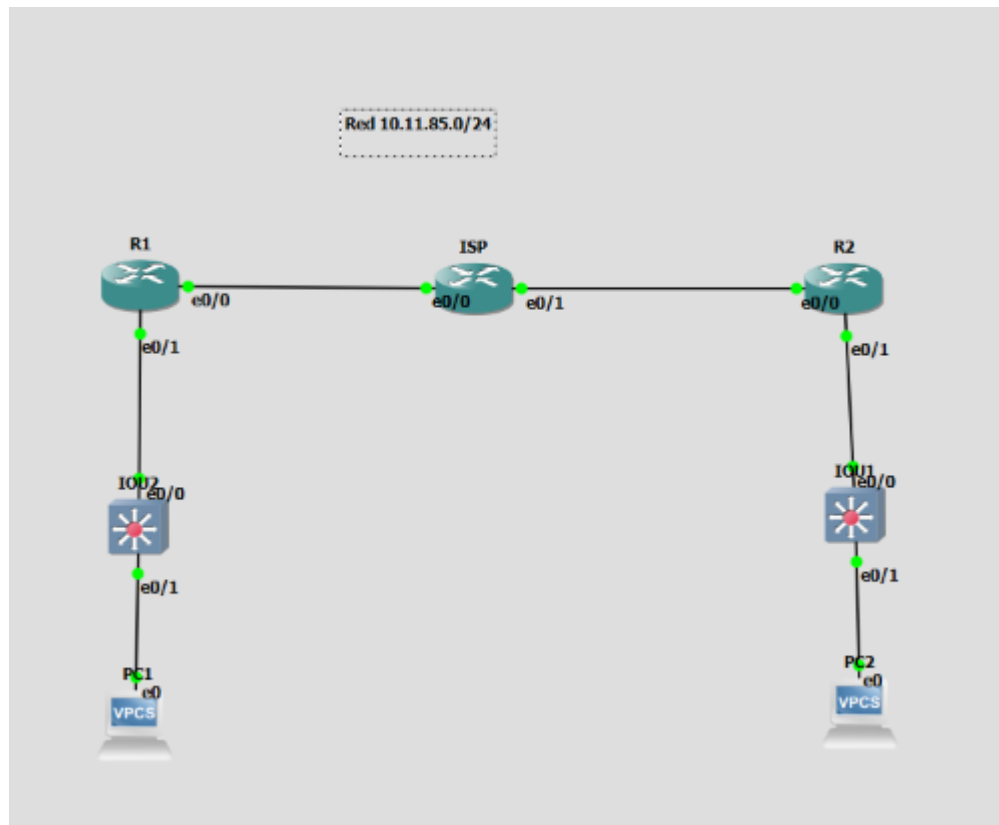


Objetivo

Implementar una VPN Site-to-Site utilizando IPSec IKEv1 con el método basado en enrutamiento (route-based) mediante una interfaz virtual de túnel (VTI — Virtual Tunnel Interface). A diferencia del enfoque policy-based, aquí no se utilizan ACLs para definir el tráfico interesante. En su lugar, se crea la interfaz lógica Tunnel0 a la cual se le aplica un IPSec Profile; cualquier paquete que el router envíe a través de dicha interfaz será cifrado automáticamente. La selección del tráfico se controla mediante la tabla de enrutamiento, lo que permite usar protocolos dinámicos como OSPF o EIGRP.

Topología de Red

Misma topología física que el lab policy-based: R1, ISP, R2, SW1, SW2, PC1, PC2. La diferencia es la adición de la interfaz lógica **Tunnel0** en R1 y R2.



Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Descripción
R1	e0/0	10.11.85.65/26	WAN → ISP
R1	e0/1	10.11.85.1/26	LAN → SW1
R1	Tunnel0	10.10.10.1/30	VTI hacia R2
ISP	e0/0	10.11.85.66/26	Hacia R1
ISP	e0/1	10.11.85.129/26	Hacia R2
R2	e0/0	10.11.85.130/26	WAN → ISP
R2	e0/1	10.11.85.193/26	LAN → SW2
R2	Tunnel0	10.10.10.2/30	VTI hacia R1
PC1	e0	10.11.85.2/26	GW: 10.11.85.1
PC2	e0	10.11.85.194/26	GW: 10.11.85.193

Parámetros IPSec IKEv1

Fase 1 — IKE / ISAKMP

Parámetro	Valor
Política	10
Cifrado	AES-256
Hash / Integridad	SHA-256
Autenticación	Pre-Shared Key (PSK)
Grupo DH	Grupo 14 (2048-bit MODP)
Lifetime	86400 segundos (24 horas)
Clave PSK	ITLA2024

Fase 2 — IPSec / Transform Set

Parámetro	Valor
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Nombre Transform Set	TS-IKEV1-VTI
Protocolo	ESP
Cifrado	AES-256
Integridad	HMAC-SHA-256
Modo	Tunnel
IPSec Profile	PROFILE-VTI

Interfaz Tunnel0

Parámetro	R1	R2
IP Tunnel	10.10.10.1/30	10.10.10.2/30
Tunnel source	Ethernet0/0	Ethernet0/0
Tunnel destination	10.11.85.130	10.11.85.65
Tunnel mode	ipsec ipv4	ipsec ipv4
Tunnel protection	PROFILE-VTI	PROFILE-VTI

Funcionamiento — Route-Based VPN

La interfaz Tunnel0 actúa como un enlace punto a punto virtual cifrado. IPSec protege automáticamente todo el tráfico que pase por ella. La decisión de cifrar un paquete no depende de ninguna ACL, sino de la tabla de enrutamiento: si la ruta hacia la LAN remota apunta a Tunnel0, el paquete se cifra. Esto permite agregar nuevas subredes simplemente añadiendo rutas, sin modificar la configuración IPSec.

PC1 (10.11.85.2) envía paquete a PC2 (10.11.85.194)

R1: route lookup → 10.11.85.192/26 via Tunnel0

Tunnel0 tiene IPSec profile → paquete se cifra con AES-256

Sale por e0/0 → ISP → R2 e0/0

R2: Tunnel0 recibe → descifra → entrega a PC2 por e0/1

■ *Ventaja clave: si se agrega la LAN 10.11.85.64/26 al sitio R1, solo se necesita una ruta adicional apuntando a Tunnel0 en R2. En policy-based habría que modificar la ACL y el crypto map.*

Scripts de Configuración

ISP

```
enable
configure terminal

hostname ISP

interface Ethernet0/0
description Hacia R1
ip address 10.11.85.66 255.255.255.192
no shutdown

interface Ethernet0/1
description Hacia R2
ip address 10.11.85.129 255.255.255.192
no shutdown

end
write memory
```

R1

```
enable
configure terminal

hostname R1
interface Ethernet0/0
description WAN hacia ISP
ip address 10.11.85.65 255.255.255.192
no shutdown

interface Ethernet0/1
description LAN hacia SW1-PC1
ip address 10.11.85.1 255.255.255.192
no shutdown

ip route 0.0.0.0 0.0.0.0 10.11.85.66

crypto isakmp policy 10
encr aes 256
hash sha256
```

*authentication pre-share
group 14
lifetime 86400*

crypto isakmp key ITLA2024 address 10.11.85.130

*crypto ipsec transform-set TS-IKEV1-VTI esp-aes 256 esp-sha256-hmac
mode tunnel*

*crypto ipsec profile PROFILE-VTI
set transform-set TS-IKEV1-VTI*

*interface Tunnel0
description VTI hacia R2
ip address 10.10.10.1 255.255.255.252
tunnel source Ethernet0/0
tunnel destination 10.11.85.130
tunnel mode ipsec ipv4
tunnel protection ipsec profile PROFILE-VTI
no shutdown*

ip route 10.11.85.192 255.255.255.192 Tunnel0

end

R2

*enable
configure terminal*

hostname R2

*interface Ethernet0/0
description WAN hacia ISP
ip address 10.11.85.130 255.255.255.192
no shutdown*

*interface Ethernet0/1
description LAN hacia SW2-PC2
ip address 10.11.85.193 255.255.255.192
no shutdown*

ip route 0.0.0.0 0.0.0.0 10.11.85.129

```
crypto isakmp policy 10
  encr aes 256
  hash sha256
  authentication pre-share
  group 14
  lifetime 86400
```

```
crypto isakmp key ITLA2024 address 10.11.85.65
```

```
crypto ipsec transform-set TS-IKEV1-VTI esp-aes 256 esp-sha256-hmac
mode tunnel
```

```
crypto ipsec profile PROFILE-VTI
set transform-set TS-IKEV1-VTI
```

```
interface Tunnel0
  description VTI hacia R1
  ip address 10.10.10.2 255.255.255.252
  tunnel source Ethernet0/0
  tunnel destination 10.11.85.65
  tunnel mode ipsec ipv4
  tunnel protection ipsec profile PROFILE-VTI
  no shutdown
```

```
ip route 10.11.85.0 255.255.255.192 Tunnel0
end
```

VPCS

```
! PC1
ip 10.11.85.2 255.255.255.192 10.11.85.1
save
```

```
! PC2
ip 10.11.85.194 255.255.255.192 10.11.85.193
save
```

Verificación

Se ejecutaron estos comandos en R1 o R2 para confirmar el funcionamiento del túnel VTI.

```
! Tunnel0 debe estar UP/UP show
interface Tunnel0
```

```
! Ruta hacia LAN remota debe apuntar a Tunnel0 show ip route !
Esperado: S 10.11.85.192/26 is directly connected, Tunnel0
```

```
! IKE Fase 1 (QM_IDLE = establecido) show
crypto isakmp sa
```

```
! IPSec Fase 2 + contadores encrypt/decrypt show
crypto ipsec sa
```

```
! Profile aplicado al túnel show
crypto ipsec profile
```

```
R1#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA

```

dst	src	state	conn-id	status
10.11.85.130	10.11.85.65	QM_IDLE	1002	ACTIVE
10.11.85.65	10.11.85.130	QM_IDLE	1001	ACTIVE

```
IPv6 Crypto ISAKMP SA
```

```
R1#sh crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.11.85.65

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.11.85.130 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 8, #pkts encrypt: 8, #pkts digest: 8
  #pkts decaps: 8, #pkts decrypt: 8, #pkts verify: 8
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

local crypto endpt.: 10.11.85.65, remote crypto endpt.: 10.11.85.130
plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb Ethernet0/0
current outbound spi: 0x197F6088(427781512)
PFS (Y/N): N, DH group: none

inbound esp sas:
  spi: 0x65898A2C(1703524908)
```

```

R1#show ip int tunnel 0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Description: VTI hacia R2
  Internet address is 10.10.10.1/30
  MTU 17878 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 10.11.85.65 (Ethernet0/0), destination 10.11.85.130
  Tunnel Subblocks:
    src-track:
      Tunnel0 source tracking subblock associated with Ethernet0/0
      Set of tunnels with source Ethernet0/0, 1 member (includes iterators), on interface <OK>
  Tunnel protocol/transport IPSEC/IP
  Tunnel TTL 255
  Tunnel transport MTU 1438 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Tunnel protection via IPsec (profile "PROFILE-VTI")
  Last input never, output never, output hang never
  Last clearing of "show interface" counters 00:25:26

R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.11.85.129 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.11.85.129
      10.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
C      10.10.10.0/30 is directly connected, Tunnel0
L      10.10.10.2/32 is directly connected, Tunnel0
S      10.11.85.0/26 is directly connected, Tunnel0
C      10.11.85.128/26 is directly connected, Ethernet0/0
L      10.11.85.130/32 is directly connected, Ethernet0/0
C      10.11.85.192/26 is directly connected, Ethernet0/1
L      10.11.85.193/32 is directly connected, Ethernet0/1

R2#
PC1 - 10.11.85.194 199.199.199.194 gateway 10.11.85.129

PC2> trace 10.11.85.2
Trace to 10.11.85.2, 8 hops max, press Ctrl+C to stop
 1  10.11.85.193  1.645 ms  7.388 ms  1.864 ms
 2  10.10.10.1   6.232 ms  1.911 ms  1.319 ms
 3  *10.11.85.2  2.431 ms (ICMP type:3, code:3, Destination port unreachable)

PC2>

```

■ A diferencia del policy-based, NO aparece 'crypto map' en 'show interface e0/0'. En cambio, 'show interface Tunnel0' muestra 'tunnel protection ipsec profile PROFILE-VTI'.